Nasser Mohammed

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University of Toronto

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iD: 0009-0008-1224-0382

Research Interests: Stellar Streams, Dark Matter, Galactic Archeology, Astrostatistics, Data Science

Education

Sep. 2024 – Present **PhD Candidate in Astronomy & Astrophysics | University of Toronto**
Thesis topic: *Indirect Dark Matter Detection via Stellar Streams*
Supervisor: Professor Ting S. Li

Sep. 2019 – Dec. 2023 **BSc Honours in Physics | University of British Columbia, Okanagan**
Minor in Data Science
Thesis Title: *Polarimetric Study of 'Tadpole' Feature G137+7 with CHIME Radio Data*
Supervisor: Professor Alex S. Hill

Publications

First Author Publications

N. Mohammed, et al. 2026, ApJ, 1005, 2: *The Kinematically Hot, Extremely Metal-Poor C-19 Stellar Stream in DESI DR2*

N. Mohammed, et al. 2024, ApJ, 971, 100: *Faraday Tomography with CHIME: The 'Tadpole' Feature G137+7*

Co-Authored Publications

E. Jarvis et al. (incl. **N. Mohammed**) 2026, submitted to ApJ: *Characterizing the GD-1 Stream with DESI DR2 Data: Thin Stream and Hot Cocoon*

A. Byström, et al. (incl. **N. Mohammed**) 2026, submitted to MNRAS: *Constraining the Galactic bar using the M92 stellar stream*

Awards & Honours

2026 NSERC Canadian Graduate Research Scholarship - Doctoral (CGSR D)
2025 NSERC Canadian Graduate Scholarship - Masters (CGS M)
2025 LSST Discovery Alliance Data Science Fellowship
2025 UofT Recognition of Excellence Award
2024 Mary Louise & Ronald Laidlaw Martin Graduate Scholarship in A&A
2024 David A. Dunlap Department of Astronomy & Astrophysics Entrance Fellowship
2024 NSERC Canadian Graduate Masters Scholarship (declined)
2024 NSERC Undergraduate Student Research Award
2023 Irving K. Barber Faculty of Science Undergraduate Research Award (2023)
2023 NSERC Undergraduate Student Research Award

Supervising & Mentoring Experience

May. 2026 – Present	Emily Moran	Research Project Co-Supervisor | Mitacs
Sep. 2025 – May 2026	Alex Brisebois	Graduate Mentorship Program | University of Toronto
Sep. 2025 – Apr. 2026	Maia Wertheim	Undergraduate Mentorship Program | University of Toronto
May 2025 – Sept. 2025	Alejandra Calderon	Research Project Co-Supervisor | Mitacs
Jan. 2025 – Oct. 2025	Isabelle Laing	Undergraduate Mentorship Program | University of Toronto
Apr. 2023 – Sept. 2023	5 Mentees	Globalink Mentorship Program | Mitacs *Mentored five international students travelling to UBC Okanagan for summer research projects.*

Leadership & Service

Upcoming, Jul. 2026	Local Organizing Committee Member: "Near Field Cosmology in the Era of Big Data: Local Group and Beyond" Workshop | The Fields Institute, University of Toronto
Oct. 2025 – Current	Union Steward | CUPE 3902 Unit 1 Union Steward for Unit 1 & Graduate Student Astronomy Association Union Representative.
Oct. 2024 – Current	Colloquium Speaker Dinner Organizer | David. A Dunlap Dept. of Astro. & Astrophysics Co-ordinates post-doc and graduate student attendees, place reservations, accomodate speaker and attendee dietary restrictions.
Oct. 2024 – Oct. 2025	Mediation Committee | Graduate Astronomy Student Association Authored report on graduate student population financial wellbeing and work-life balance. Successfully advocated for increase in astronomy graduate student stipends.
May 2023 – May 2024	Co-Founder & President of the UBC Okanagan Astronomy Club Hosted seminars, coordinated stargazing nights, organized youth STEM outreach initiatives.

Further Education & Select Attendances

Feb. 2025 – Present	LSST Discovery Alliance Data Science Fellowship Program
Feb 2026	RE:SciComm - Canada's Science Communication Conference
May 2023	Canadian Astroparticle Physics Summer School, Queen's University & SNOLAB

Media Appearances

June 2025	*Telescope images give 1st ultra HD images of the universe* | CBC News On-camera interview; clip included in aired segment

Observing Experience

26 Sep. 2025 – 26 Sep. 2025	Co-Investigator, Las Campanas Observatory | Región de Coquimbo, Chile 2 nights on Magellan/MIKE; Observations of the C-19 stellar stream

List of Talks & Lectures

May 2026	AST101 Substitute Lecturer | University of Toronto 2 hr Lecture
Oct. 2025	IAU Symposium 403: Hidden Beauty of the Galactic Outskirts | Córdoba, Spain Oral presentation
Oct. 2025	AST101 Substitute Lecturer | University of Toronto 1 hr Lecture
Aug. 2025	York University EXPLORE Summer School | North York, ON, Canada Lecturer
Jun. 2025	Clusters in Halifax Workshop | Halifax, NS, Canada 15-minute talk
Jun. 2025	CASCA Annual General Meeting 2025 | Halifax, NS, Canada Poster presentation
Mar. 2024	Okanagan College | Kelowna, BC, Canada Guest lecture: Introduction to the Interstellar Medium (ASTR 121)
Feb. 2024	University of Calgary PHAS Symposium | Calgary, AB, Canada 15-minute talk

Sep. 2023 UBC Okanagan Faculty of Science URA Symposium | Kelowna, BC, Canada
Poster presentation

Jun. 2023 CASCA Annual General Meeting 2023 | Penticton, BC, Canada
Poster presentation

Jun. 2023 TomFest: Frontiers in Radio Science and Engineering | Kaleden, BC, Canada
15-minute talk